



BigBasket

GROCERY FORECAST

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EDA with Big Basket Data

E-COMMERCE (ELECTRONIC COMMERCE) IS THE ACTIVITY OF ELECTRONICALLY BUYING OR SELLING OF PRODUCTS ON ONLINE SERVICES OR OVER THE INTERNET. E- COMMERCE DRAWS ON TECHNOLOGIES SUCH AS MOBILE COMMERCE, ELECTRONIC FUNDS TRANSFER, SUPPLY CHAIN MANAGEMENT, INTERNET MARKETING, ONLINE TRANSACTION PROCESSING, ELECTRONIC DATA INTERCHANGE (EDI), INVENTORY MANAGEMENT SYSTEMS, AND AUTOMATED DATA COLLECTION SYSTEMS. E- COMMERCE IS IN TURN DRIVEN BY THE TECHNOLOGICAL ADVANCES OF THE SEMICONDUCTOR INDUSTRY, AND IS THE LARGEST SECTOR OF THE ELECTRONICS INDUSTRY. EDA WITH BIG BASKET DATA BIG BASKET IS THE LARGEST ONLINE GROCERY SUPERMARKET IN INDIA. WAS LAUNCHED SOMEWHERE AROUND IN 2011 SINCE THEN THEY HAVE BEEN EXPANDING THEIR BUSINESS. THOUGH SOME NEW COMPETITORS HAVE BEEN ABLE TO SET THEIR FOOT IN THE NATION SUCH AS BLINK IT ETC. BUT BIG BASKET HAS STILL NOT LOST ANYTHING – THANKS TO EVER EXPANDING POPULAR BASE AND THEIR SHIFT TO ONLINE BUYING.

This dataset contains 10 attributes with simple meaning and which are described as follows:

INDEX – SIMPLY THE INDEX!

PRODUCT – TITLE OF THE PRODUCT (AS THEY'RE LISTED)

CATEGORY – CATEGORY INTO WHICH PRODUCT HAS BEEN CLASSIFIED

SUB_CATEGORY – SUBCATEGORY INTO WHICH PRODUCT HAS BEEN KEPT

BRAND – BRAND OF THE PRODUCT

SALE_PRICE – PRICE AT WHICH PRODUCT IS BEING SOLD ON THE SITE

MARKET_PRICE – MARKET PRICE OF THE PRODUCT

TYPE – TYPE INTO WHICH PRODUCT FALLS

RATING – RATING THE PRODUCT HAS GOT FROM ITS CONSUMERS

DESCRIPTION – DESCRIPTION OF THE DATASET (IN DETAIL)

Steps to Follow

STEP 1: LOAD DATASET.

STEP 2: USE HEAD FUNCTION TO LOOK FOR FIRST 12 ROWS.

STEP 3: GET DESCRIPTION OF THE DATA IN THE DATAFRAME

STEP 4: FIND INFORMATION ABOUT THE DATAFRAME.

STEP 5: FIND OUT TOP & LEAST SOLD PRODUCTS.

STEP 6: MEASURING DISCOUNT ON A CERTAIN ITEM.

**STEP 7: FIND OUT THE MISSING VALUES FROM
THE DATASET.**

**STEP 8: FIND OUT THE OUTLIERS FROM THE DATASET
ACCORDING TO THE COLUMNS AND FILL THEM WITH
THE MEAN.**

STEP 9: CREATE PLOTS OR VISUALIZATIONS.

Libraries we can used with this EDA process:



matplotlib



seaborn

Import some Library

```
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
import numpy as np
```

We Import Drive from Google Collab

```
from google.colab import drive
drive.mount("/content/drive")
```

Then we Perform Steps

STEP 1: LOAD DATASET

```
x="/content/drive/MyDrive/BigBasket Products.csv"
df=pd.read_csv(x)
df
```

0	1	Garlic Oil - Vegetarian Capsule 500 mg	Beauty & Hygiene	Hair Care	Sri Sri Ayurveda	220.00	220.0	Hair Oil & Serum	4.1
1	2	Water Bottle - Orange	Kitchen, Garden & Pets	Storage & Accessories	Mastercook	180.00	180.0	Water & Fridge Bottles	2.3
2	3	Brass Angle Deep - Plain, No.2	Cleaning & Household	Pooja Needs	Trm	119.00	250.0	Lamp & Lamp Oil	3.4
3	4	Cereal Flip Lid Container/Storage Jar - Assort...	Cleaning & Household	Bins & Bathroom Ware	Nakoda	149.00	176.0	Laundry, Storage Baskets	3.7
4	5	Creame Soft Soap - For Hands & Body	Beauty & Hygiene	Bath & Hand Wash	Nivea	162.00	162.0	Bathing Bars & Soaps	4.4
...	...	...	...	...	...	...	...	...	...
27550	27551	Wottagirl! Perfume Spray - Heaven, Classic	Beauty & Hygiene	Fragrances & Deos	Layerr	199.20	249.0	Perfume	3.9
27551	27552	Rosemary	Gourmet & World Food	Cooking & Baking Needs	Puramate	67.50	75.0	Herbs, Seasonings & Rubs	4.0
...	...	Peri-Peri Sweet Potato	Gourmet &	Snacks, Dry	...	...	...	...	...

Step 2: Use head function to look for first 12 rows.



```
df.head(12)
```

index		product	category
0	1	Garlic Oil - Vegetarian Capsule 500 mg	Beauty & Hygiene
1	2	Water Bottle - Orange	Kitchen, Garden & Pets
2	3	Brass Angle Deep - Plain, No.2	Cleaning & Household
3	4	Cereal Flip Lid Container/Storage Jar - Assort...	Cleaning & Household
4	5	Creme Soft Soap - For Hands & Body	Beauty & Hygiene
5	6	Germ - Removal Multipurpose Wipes	Cleaning & Household
6	7	Multani Mati	Beauty & Hygiene
7	8	Hand Sanitizer - 70% Alcohol Base	Beauty & Hygiene
8	9	Biotin & Collagen Volumizing Hair Shampoo + Bi...	Beauty & Hygiene
9	10	Scrub Pad - Anti- Bacterial, Regular	Cleaning & Household
10	11	Wheat Grass Powder - Raw	Gourmet & World Food
11	12	Butter Cookies Gold Collection	Gourmet & World Food

Step 3: Get Description of the data in the DataFrame



```
df.describe()
```

	index	sale_price	market_price	rating
count	27555.00000	27549.000000	27555.000000	18919.000000
mean	13778.00000	334.648391	382.056664	3.943295
std	7954.58767	1202.102113	581.730717	0.739217
min	1.00000	2.450000	3.000000	1.000000
25%	6889.50000	95.000000	100.000000	3.700000
50%	13778.00000	190.320000	220.000000	4.100000
75%	20666.50000	359.000000	425.000000	4.300000
max	27555.00000	112475.000000	12500.000000	5.000000

STEP 4: FIND INFORM ATION ABOUT THE DATAFRAME.



```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 27555 entries, 0 to 27554
Data columns (total 10 columns):
 #   Column                Non-Null Count  Dtype  
---  -
 0   index                 27555 non-null  int64  
 1   product               27554 non-null  object  
 2   category              27555 non-null  object  
 3   sub_category          27555 non-null  object  
 4   brand                 27554 non-null  object  
 5   sale_price            27549 non-null  float64 
 6   market_price          27555 non-null  float64 
 7   type                  27555 non-null  object  
 8   rating                18919 non-null  float64 
 9   description            27440 non-null  object  
dtypes: float64(3), int64(1), object(6)
memory usage: 2.1+ MB
```

Step 5: Find out Top & least sold products.

```
▶ # Group by product and sum sale_price
product_sales = df.groupby('product')['sale_price'].sum()

# Top selling product
top_product = product_sales.sort_values(ascending=False)

# Least selling product
least_product = product_sales.sort_values(ascending=True)

print("Top Selling Products:")
print(top_product.head())

print("Least Selling Products:")
print(least_product.head())
```

#GROUP BY PRODUCT AND SUM SALE_PRICE

**TOP SELLING PRODUCT=USE SORT_VALUES ON VARIABLE TO
FIND OUT TOP SELLING PRODUCT AND LEAST SELLING PRODUCT**

Top Selling Products:

product	
Beard Kit	112475.00
4mm Aluminium Induction Base Chapati Roti Tawa - Silver	112178.00
Balloon - Polka Dot, 12 Inch	88899.00
Extra Virgin Olive Oil	24808.53
Olive Oil - Extra Virgin	22568.22

Name: sale_price, dtype: float64

Least Selling Products:

product	
Puja Flower Wicks - Puvvu Vathulu Batti	0.0
Steel Fork - Medium, Premium Excel Series, BBCL08	0.0
Klassic Plain Cocktail Napkins (22 X 22 cm)	0.0
Snack Mix - Dhokla	0.0
Battery AA 3UT Hi Top	0.0

Name: sale_price, dtype: float64

Step 6: Measuring discount on a certain item.

```
df['sale_price'] = pd.to_numeric(df['sale_price'])
df['discount'] = df['market_price'] - df['sale_price']
df['discount_percent'] = (df['discount'] / df['market_price']) * 100
print(df)
```

**WE MEASURING DISCOUNT
ON MARKET PRICE AND
SALE PRICE
WE USE
PD.TO_NUMERIC
BECAUSE THIS FUNCTION
WORK AS TO CONVERT AN
ARGUMENT (SUCH AS A
SERIES, LIST) INTO A
NUMERIC DATA TYPE**

	index	product \
0	1	Garlic Oil - Vegetarian Capsule 500 mg
1	2	Water Bottle - Orange
2	3	Brass Angle Deep - Plain, No.2
3	4	Cereal Flip Lid Container/Storage Jar - Assort...
4	5	Creame Soft Soap - For Hands & Body
...	...	...
27550	27551	Wottagirl! Perfume Spray - Heaven, Classic
27551	27552	Rosemary
27552	27553	Peri-Peri Sweet Potato Chips
27553	27554	Green Tea - Pure Original
27554	27555	United Dreams Go Far Deodorant

	category	sub_category \
0	Beauty & Hygiene	Hair Care
1	Kitchen, Garden & Pets	Storage & Accessories
2	Cleaning & Household	Pooja Needs
3	Cleaning & Household	Bins & Bathroom Ware
4	Beauty & Hygiene	Bath & Hand Wash
...	...	...
27550	Beauty & Hygiene	Fragrances & Deos
27551	Gourmet & World Food	Cooking & Baking Needs
27552	Gourmet & World Food	Snacks, Dry Fruits, Nuts
27553	Beverages	Tea
27554	Beauty & Hygiene	Men's Grooming

	brand	sale_price	market_price \
0	Sri Sri Ayurveda	220.00	220.0
1	Mastercook	180.00	180.0
2	Trm	119.00	250.0
3	Nakoda	149.00	176.0
4	Nivea	162.00	162.0
...	...	...	...
27550	Layerr	199.20	249.0
27551	Puramate	67.50	75.0
27552	FabBox	200.00	200.0
27553	Tetley	396.00	495.0
27554	United Colors Of Benetton	214.53	390.0

	type	rating	discount	discount_percent
0	Hair Oil & Serum	4.1	0.00	0.000000
1	Water & Fridge Bottles	2.3	0.00	0.000000
2	Lamp & Lamp Oil	3.4	131.00	52.400000
3	Laundry, Storage Baskets	3.7	27.00	15.340909
4	Bathing Bars & Soaps	4.4	0.00	0.000000
...	...	...	...	...
27550	Perfume	3.9	49.80	20.000000
27551	Herbs, Seasonings & Rubs	4.0	7.50	10.000000
27552	Nachos & Chips	3.8	0.00	0.000000
27553	Tea Bags	4.2	99.00	20.000000
27554	Men's Deodorants	4.5	175.47	44.992308

Step 7: Find out the Missing Values from the Dataset.

**FIRST OF ALL WE CLEANING THE DATA
WE DROP THE COLUMN AND ROW THERE
ARE MISSING VALUES ABOVE 70%**

```
df.drop("description",axis=1,inplace=True)
```

THEN WE LOAD THE DATA

```
list(df.columns)
```

```
['index',  
 'product',  
 'category',  
 'sub_category',  
 'brand',  
 'sale_price',  
 'market_price',  
 'type',  
 'rating']
```

Searching how many null values are there



```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 27555 entries, 0 to 27554
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   index                 27555 non-null  int64
1   product               27554 non-null  object
2   category              27555 non-null  object
3   sub_category          27555 non-null  object
4   brand                 27554 non-null  object
5   sale_price            27549 non-null  float64
6   market_price          27555 non-null  float64
7   type                  27555 non-null  object
8   rating                18919 non-null  float64
dtypes: float64(3), int64(1), object(5)
memory usage: 1.9+ MB
```

STEP 8: FIND OUT THE OUTLIERS FROM THE DATASET ACCORDING TO THE COLUMNS AND FILL THEM WITH THE MEAN.

Four columns contain null values.

In this column we use Mode (Aggregate Function) because the column is Object Type



```
df["product"].mode()
```

product

0 Turmeric Powder/Arisina Pudi

dtype: object

After Findout Mode value Fill them with the values(Mode)

.fillna(method in pandas is used to replace missing (NaN/NULL) values in a DataFrame or Series with specific data)



```
df["product"].fillna(" Turmeric Powder/Arisina Pudi",inplace=True)
```

In this column we use Mode (Aggregate Function)
because the column is Object Type

```
df["brand"].mode()
```

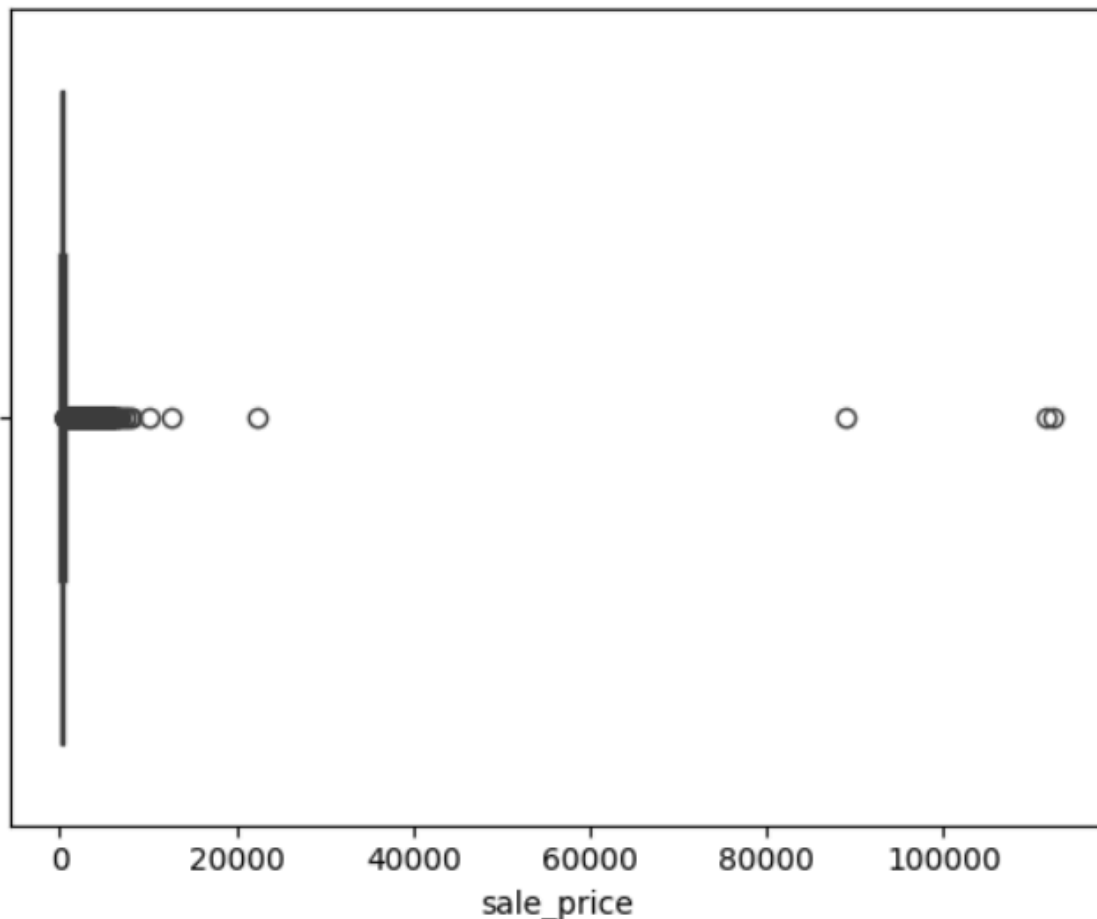
```
brand
0    Fresho
dtype: object
```

```
df["brand"].fillna("Fresho",inplace=True)
```

**MAKE A SEABORN BOXPLOT TO SEE HOW MANY
OUTLIERS ARE THERE**

```
sns.boxplot(data=df,x="sale_price")
```

<Axes: xlabel='sale_price'>



Find out Outliers from sale price by using IQR method

```

a=df["sale_price"]
q1=a.quantile(0.25)
q3=a.quantile(0.75)
iqr=q3-q1
lower_limit=q1-(1.5*iqr)
upper_limit=q3+(1.5*iqr)
print("upper limit",upper_limit)
print('lower_limit',lower_limit)
outliers=df[(a< lower_limit) | (a >upper_limit)]
print("outliers",outliers)

```

```

upper limit 755.0
lower_limit -301.0
outliers      index      product \
8            9  Biotin & Collagen Volumizing Hair Shampoo + Bi...
47           48                Colour Catcher Sheets
51           52                Peach Syrup
69           70      Padded Harness - 3/4 inch, Grey Colour
91           92  Hard Anodised Ezee-Pour Saucepan With Lid - L88
...          ...
27498  27499      Juicer - Fruit & Vegetable, Light Green
27505  27506      Virgin Coconut Oil
27514  27515      Verge & Sheer Perfume For Pair
27538  27539  Quista Pro Advanced Whey Protein Formula forti...
27542  27543      Embellish Skin Lightening Serum

      category      sub_category      brand \
8      Beauty & Hygiene      Hair Care      StBotanica
47      Cleaning & Household      All Purpose Cleaners      Dylan
51      Gourmet & World Food      Drinks & Beverages      Pekers
69      Kitchen, Garden & Pets      Pet Food & Accessories      Glenand
91      Kitchen, Garden & Pets      Cookware & Non Stick      Hawkins Futura
...      ...
27498      Kitchen, Garden & Pets      Kitchen Accessories      Ganesh
27505      Foodgrains, Oil & Masala      Edible Oils & Ghee      Merkera
27514      Beauty & Hygiene      Fragrances & Deos      Skinn by Titan
27538      Beauty & Hygiene      Health & Medicine      Himalaya
27542      Beauty & Hygiene      Skin Care      Organic Harvest

      sale_price  market_price      type  rating
8      1098.00      1098.0      Shampoo & Conditioner      3.5
47      799.00      799.0      Imported Cleaners      4.0
51      850.00      850.0      Gourmet Juices & Drinks      4.2
69      840.00      840.0      Pet Collars & Leashes      NaN
91      864.50      910.0      Tawa & Sauce Pan      4.6
...      ...
27498      1071.00      1071.0      Kitchen Tools & Other Accessories      2.0
27505      875.00      875.0      Other Edible Oils      NaN
27514      1615.50      1795.0      Perfume      NaN
27538      4500.00      4500.0      Supplements & Proteins      4.0
27542      1525.75      1795.0      Face Care      4.2

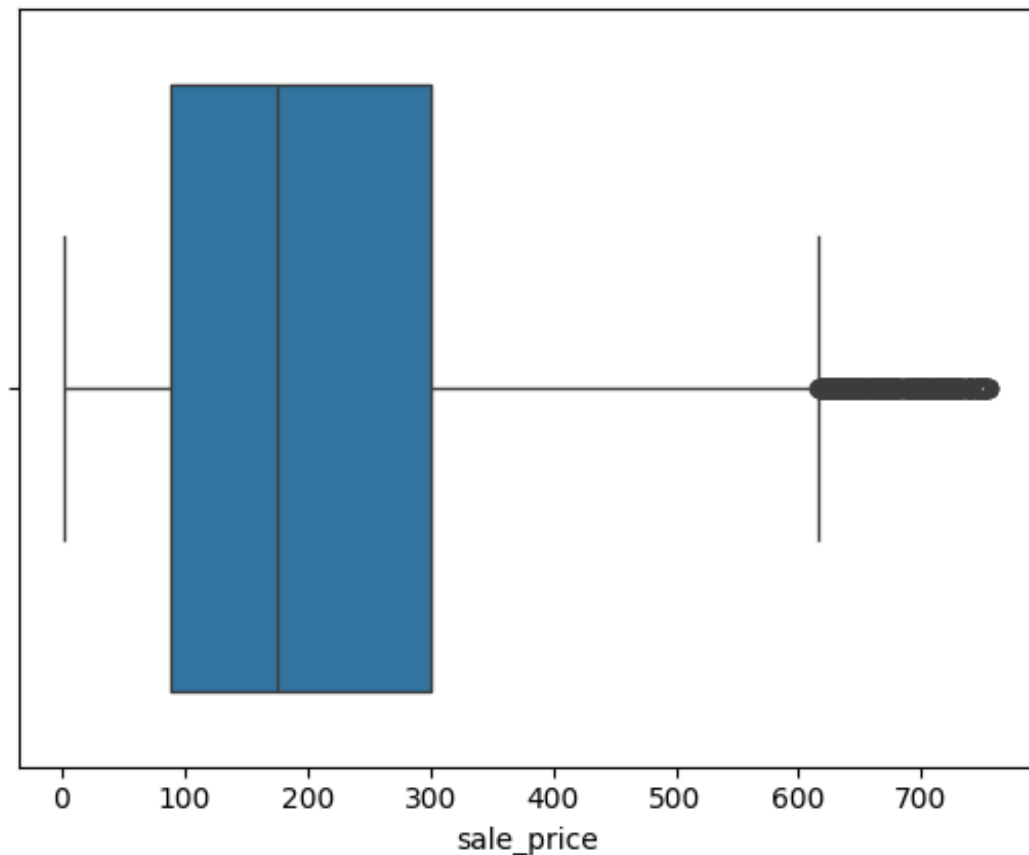
```

Remove the Outliers is far from the Q1(LOWER FENCE) AND Q3(UPPER FENCE)

```
df_clean=df[(a>=lower_limit) & (a<=upper_limit)]
```

```
sns.boxplot(x=df_clean['sale_price'])
```

```
... <Axes: xlabel='sale_price'>
```



**AFTER REMOVING
OUTLIERS THE NULL
VALUES FILLING WITH
MEAN**

```
df['sale_price'].mean()
```

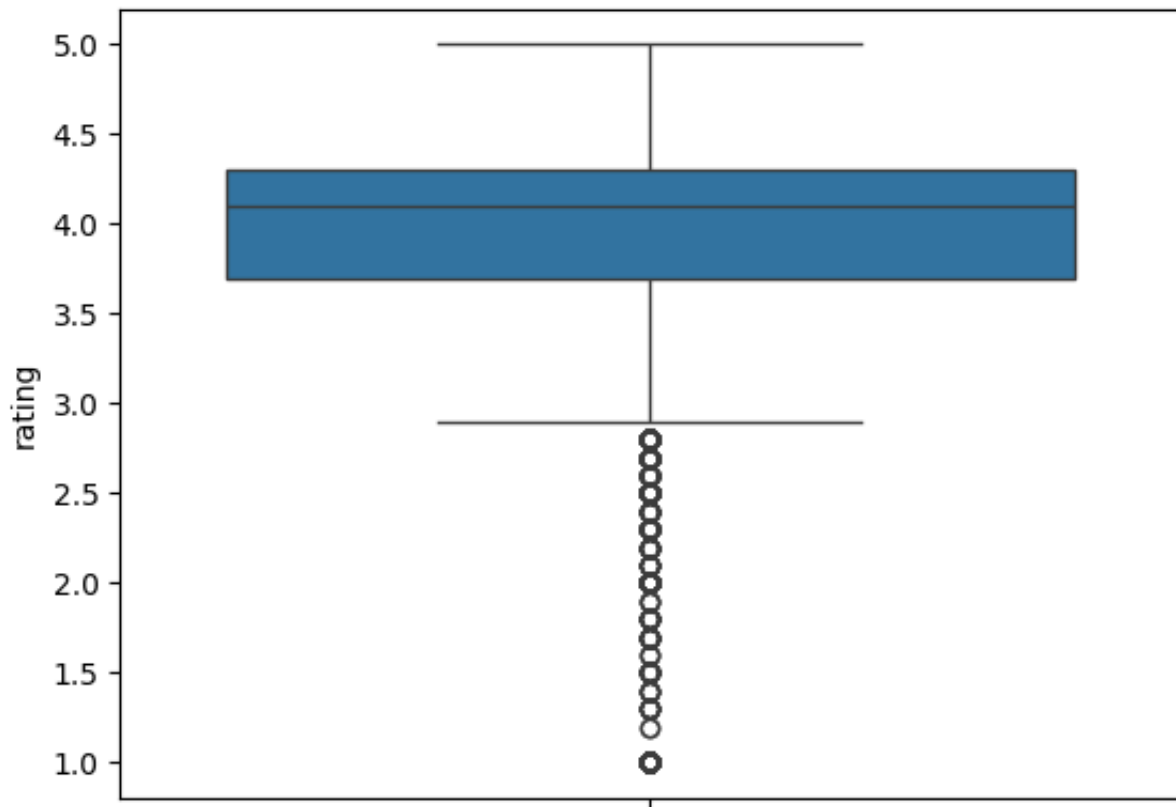
```
... np.float64(334.6483912301717)
```

```
df['sale_price'].fillna('334.6483912301717',inplace=True)
```

Make Boxplot to see how many Outliers are there

```
sns.boxplot(data=df,y="rating")
```

... <Axes: ylabel='rating'>



FIND OUT OUTLIERS FROM RATING BY USING IQR METHOD

```
#find out (RATING) outliers
y=df['rating']

q1=y.quantile(0.25)
q3=y.quantile(0.75)
iqr=q3-q1

lower_limit=q1-(1.5*iqr)
upper_limit=q3+(0.75*iqr)
print("lower_limit",lower_limit)
print("upper_limit",upper_limit)

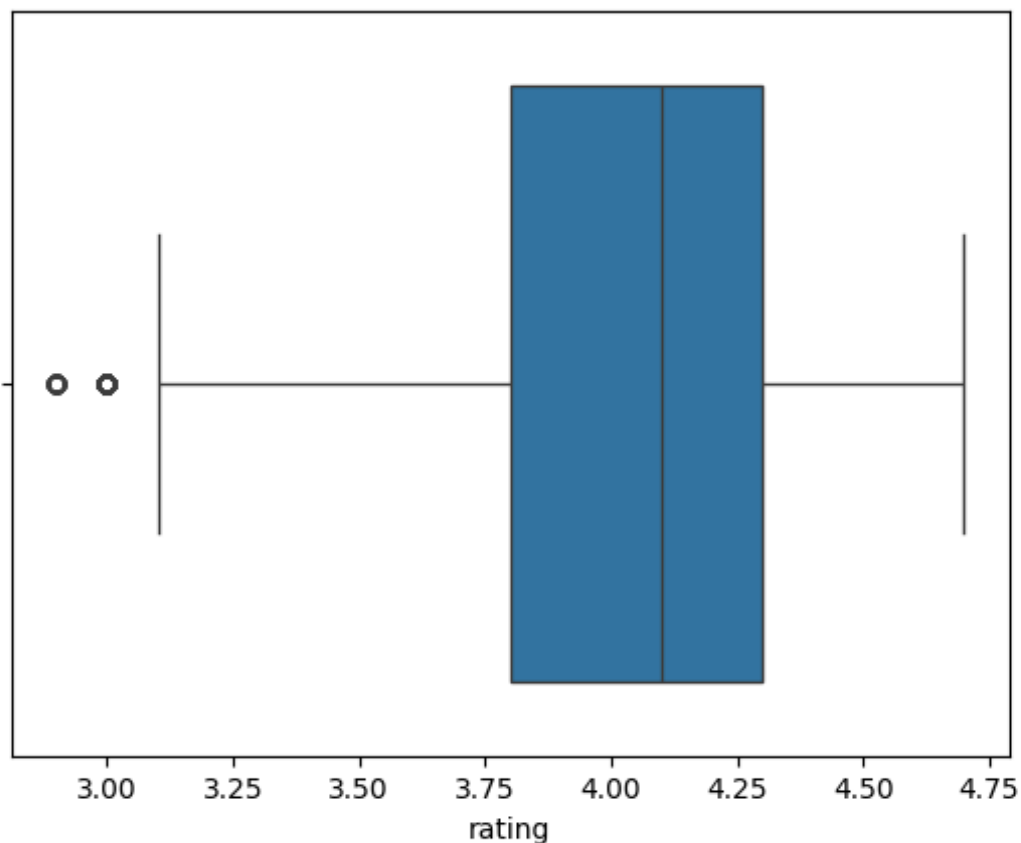
outliers=[(y > upper_limit) | (y < lower_limit)]
print("outliers",outliers)
```

```
lower_limit 2.8000000000000007
upper_limit 4.75
outliers [0      True
1       True
2       True
3       True
4       True
...
27550   True
27551   True
27552   True
27553   True
27554   True
Name: rating, Length: 27555, dtype: bool]
```

*Remove the Outliers is far from the
Q1(LOWER FENCE) AND Q3(UPPER
FENCE)*

```
sns.boxplot(data=df_clean2,x="rating")
```

... <Axes: xlabel='rating'>



**AFTER REMOVING
OUTLIERS THE NULL
VALUES FILLING WITH
MEAN**

```
df['rating'].mean()  
... np.float64(3.9432951001638563)
```

```
df['rating'].fillna('3.9432951001638563',inplace=True)
```

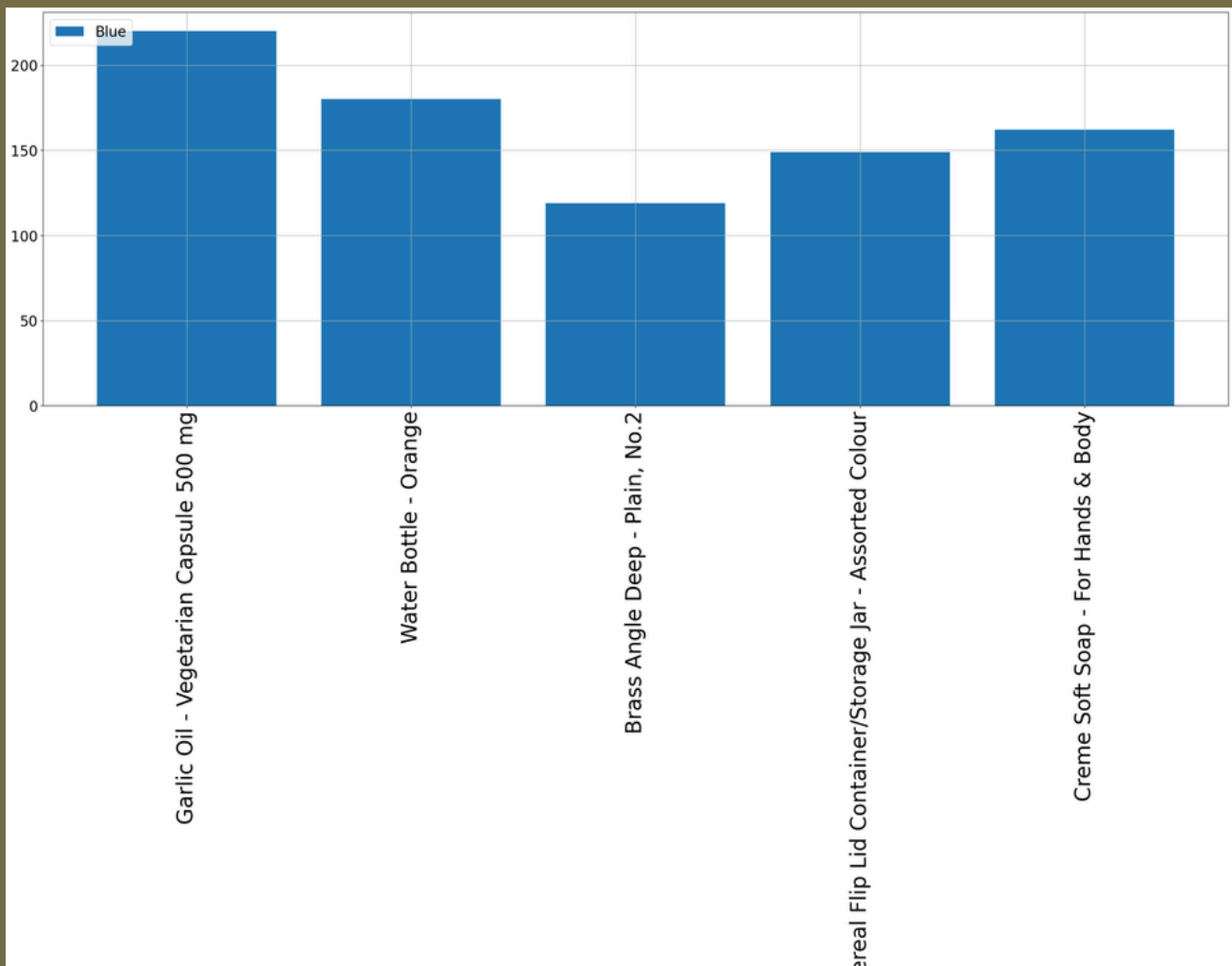
**AFTER
REMOVING ALL
THE OUTLIERS
AND FILL THE
NULL VALUES
WITH MEAN**

```
df.info()  
... <class 'pandas.core.frame.DataFrame'>  
RangeIndex: 27555 entries, 0 to 27554  
Data columns (total 9 columns):  
#   Column                Non-Null Count  Dtype  
---  -  
0   index                  27555 non-null  int64  
1   product                27555 non-null  object  
2   category               27555 non-null  object  
3   sub_category           27555 non-null  object  
4   brand                  27555 non-null  object  
5   sale_price             27555 non-null  object  
6   market_price           27555 non-null  float64  
7   type                   27555 non-null  object  
8   rating                 27555 non-null  object  
dtypes: float64(1), int64(1), object(7)  
memory usage: 1.9+ MB
```

Step 9: Create Plots or visualizations.

#Making Matplotlib Bar

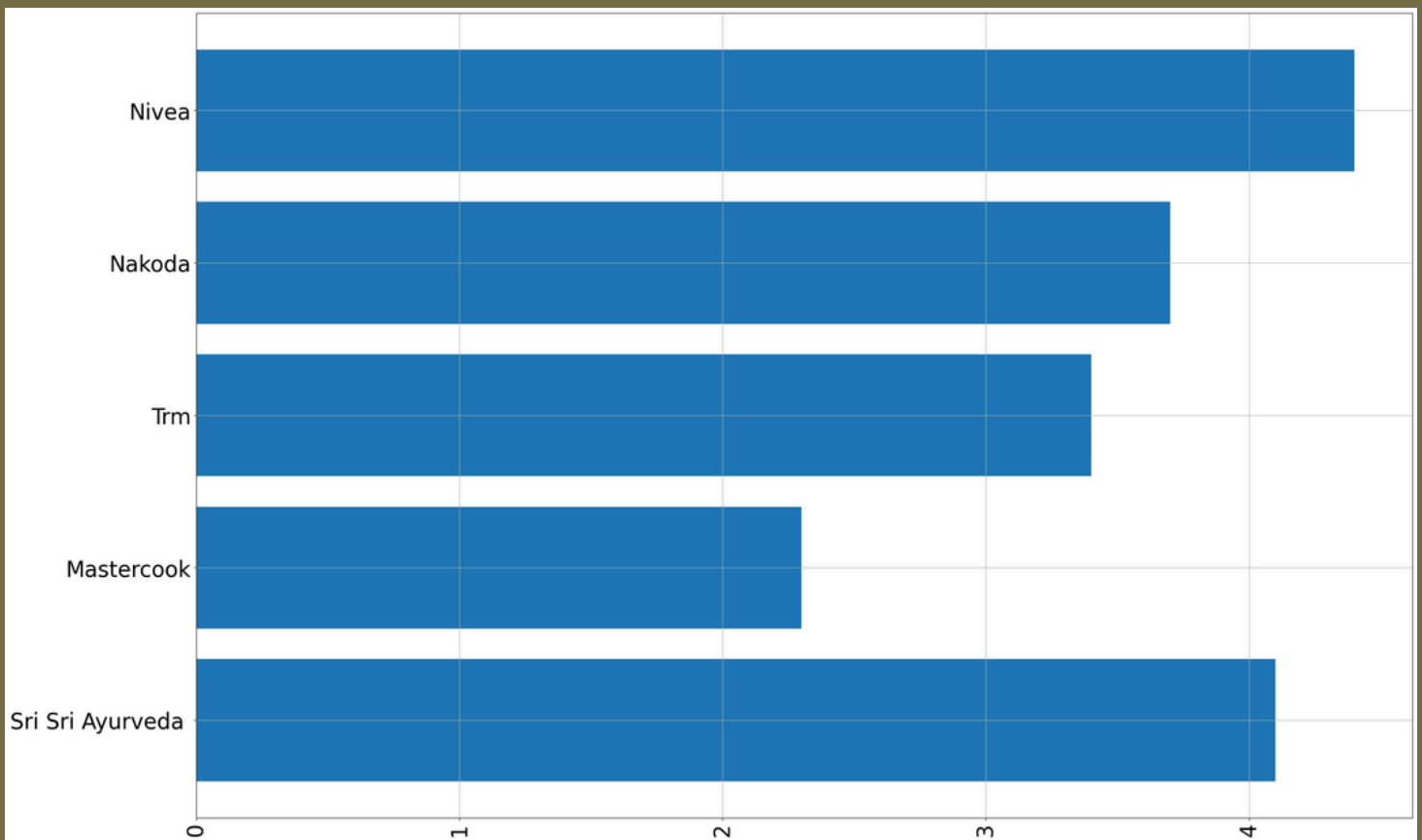
```
plt.figure(figsize=(30,10))  
plt.xticks(fontsize=30)  
plt.yticks(fontsize=20)  
plt.bar(z["product"],z["sale_price"],label="Blue")  
plt.legend(loc="upper left",fontsize=20)  
plt.xticks(rotation=90)  
plt.grid()  
plt.show()
```



THE GARLIC OIL PRODUCT SIGNIFICANTLY OUTPERFORMS THE OTHERS, WHILE THE BRASS ANGLE ITEM LAGS BEHIND. THE REMAINING THREE PRODUCTS SHOW MODERATE AND RELATIVELY CLOSE PERFORMANCE, FORMING A MIDDLE TIER.

#Making Matplotlib #horizontal Bar

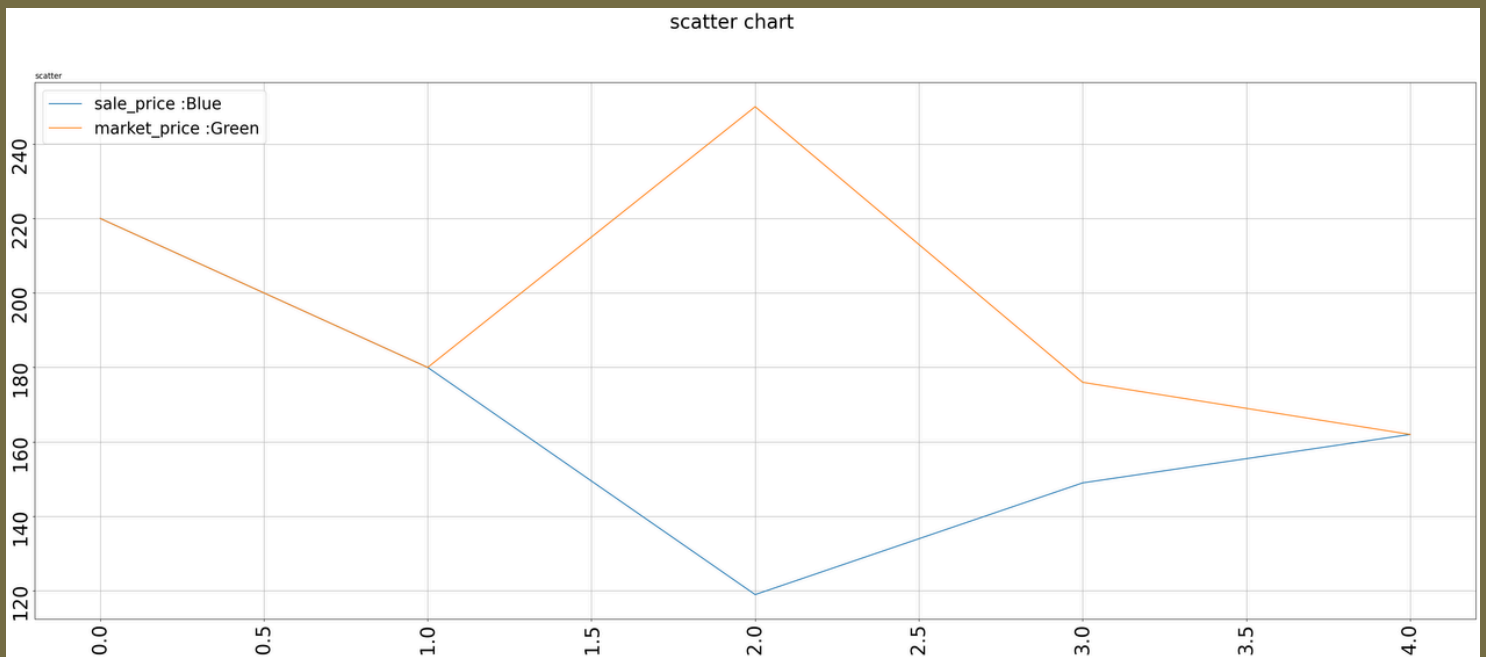
```
▶ plt.figure(figsize=(30,20))  
plt.barh(z["brand"],z["rating"])  
plt.xticks(rotation=90,fontsize=30)  
plt.yticks(fontsize=30)  
plt.grid()  
plt.show()
```



**THE CHART SHOWS A CLEAR RANKING OF BRANDS BASED ON THEIR RATING
NIVEA RATING IS FAR BETTER THAN OTHER PRODUCTS LIKE NAKODA , TRM ,
MASTERCOOK , SRI SRI AYURVEDA .
MASTERCOOK HAS THE LOWEST RATING, INDICATING COMPARATIVELY WEAKER
STANDING.**

#Making Matplotlib Scatter Plot

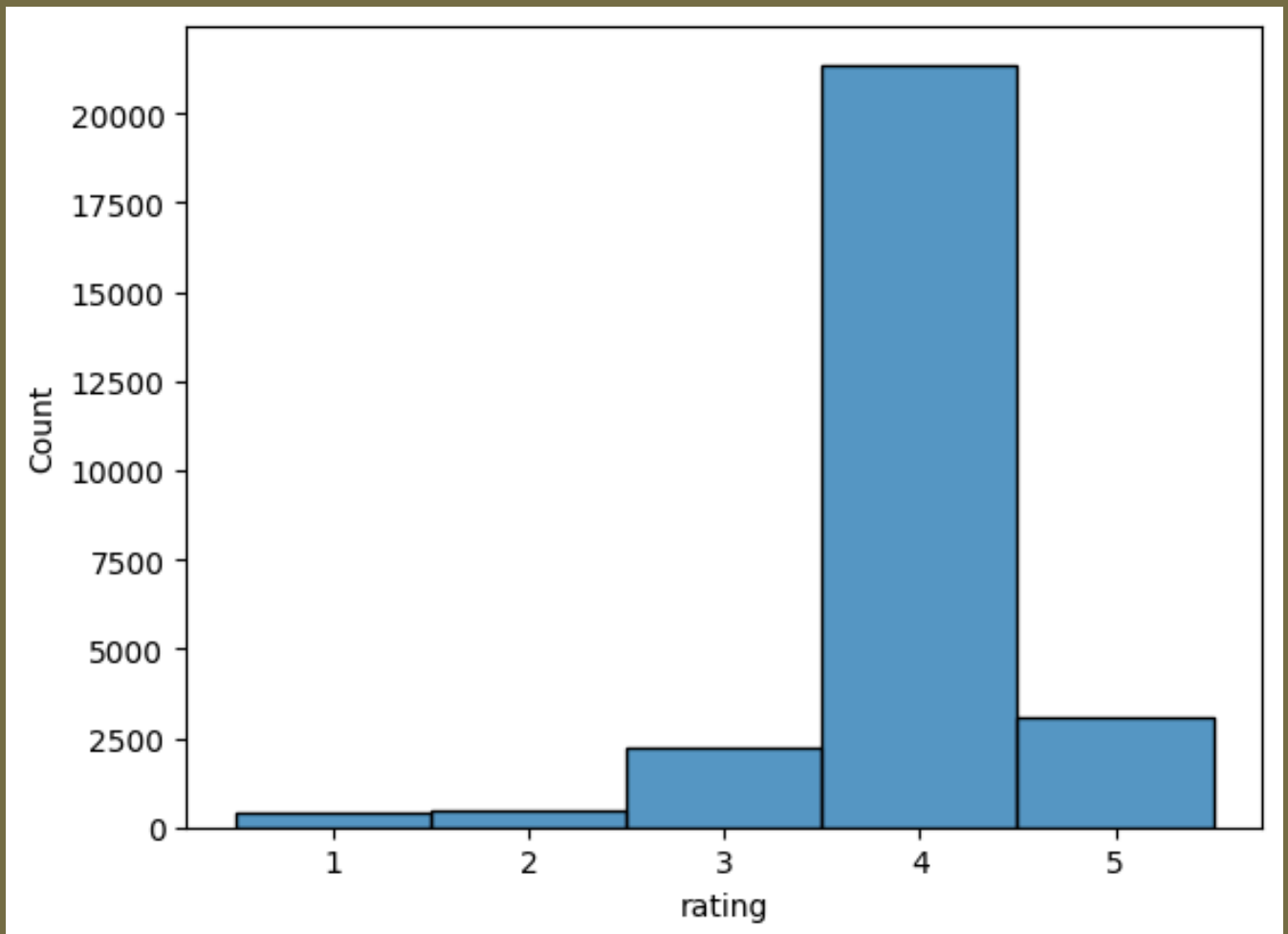
```
plt.figure(figsize=(40,15))
plt.plot(z["sale_price"],label="sale_price :Blue",)
plt.plot(z["market_price"],label="market_price :Green")
plt.legend(loc="upper left",fontsize=25)
plt.xticks(rotation=90,fontsize=30)
plt.yticks(rotation=90,fontsize=30)
plt.title("scatter" , loc="left")
plt.suptitle('scatter chart',fontsize=30)
plt.grid()
plt.show()
```



THE CHART SHOWS A CONSISTENT GAP BETWEEN MARKET PRICE AND SALE PRICE ACROSS MOST POINTS, WITH THE MARKET PRICE GENERALLY REMAINING HIGHER. THE LARGEST DIFFERENCE OCCURS AROUND THE THIRD DATA POINT, WHERE THE MARKET PRICE PEAKS WHILE THE SALE PRICE DROPS SIGNIFICANTLY, INDICATING A STRONG DISCOUNT OR PRICING GAP AT THAT STAGE.

#Making Seaborn Histogram

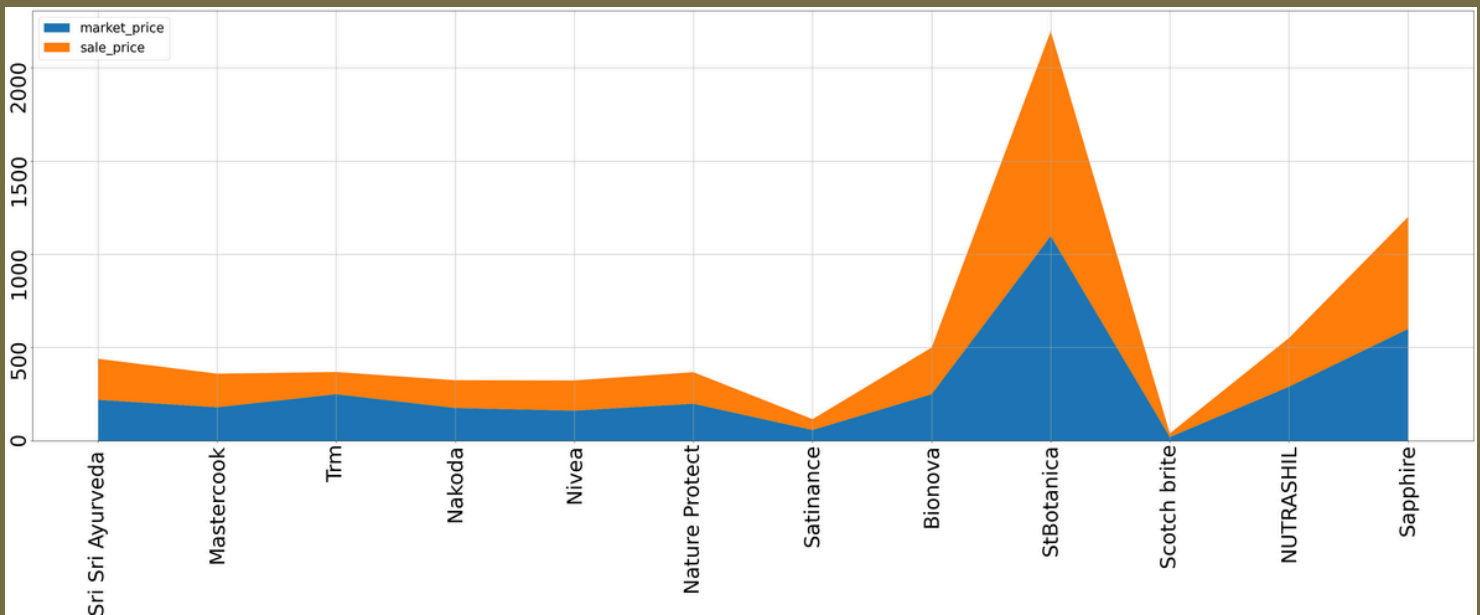
```
▶ histogram=sns.histplot(x="rating",data=df)
```



MOST OF THE RATINGS ARE HIGH, ESPECIALLY AROUND 4. THIS MEANS PEOPLE ARE GENERALLY HAPPY OR SATISFIED. VERY FEW PEOPLE GAVE LOW RATINGS LIKE 1 OR 2. SOME GAVE A 3, BUT NOT AS MANY AS 4. THERE ARE ALSO SOME 5 RATINGS, SHOWING VERY HIGH SATISFACTION. OVERALL, THE FEEDBACK IS MOSTLY POSITIVE.

#Making Seaborn Stackplot

```
plt.figure(figsize=(50,15))
plt.stackplot(z['brand'],z['market_price'],z['sale_price'],labels=["market_price","sale_price"])
plt.legend(loc="upper left",fontsize=25)
plt.xticks(rotation=90,fontsize=40)
plt.yticks(rotation=90,fontsize=40)
plt.grid()
plt.show()
```



- Most brands offer noticeable discounts, as the sale price is consistently lower than the market price.
- Brands like StBotanica and Sapphire have the highest price values, indicating premium positioning and larger discount margins.
- Some brands such as Satinance and Scotch Brite show very low pricing, suggesting budget-friendly options.
- Mid-range brands like Nivea, Biotique, and Mamaearth maintain moderate pricing with reasonable discounts.

Conclusion

1. Easy to order groceries, electronics items, kitchen items and so on online.
2. The wide variety of products, good quality of fresh items, and the comfort of home delivery, which helps save time and effort.
3. It also offers discounts that make shopping more affordable.
4. Deliveries can sometimes be delayed, certain items may be out of stock, and prices can be a bit higher compared to local stores.
5. Customers also face issues with product quality.
6. BigBasket stands out as a reliable platform that is transforming the way people shop for groceries by making the process faster, simpler, and more convenient.